



Derya Sarikaya, Dipl.-Ing. (FH)

AI Solutions Engineer | Technical AI Consultant

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AI Solutions Engineer with hands-on experience in Python, FastAPI, backend development, LLM applications, and RAG-based systems, combined with 10+ years of engineering and technical consulting experience within the Mercedes-Benz environment. Strong ability to identify patterns, structure complex technical problems, and translate business and stakeholder requirements into practical solutions.

Technical Skills

AI & Software Engineering

RAG Systems • AI Agents & Orchestration • LLM Integration • GenAI

Backend Development

Python • FastAPI • PostgreSQL • REST API Architecture • Unit Testing

Process & Engineering

Business Process Optimization • Requirements Engineering • API Design

DevOps & Tools

Git & GitHub

Soft Skills

Analytical Problem Solving • Systems Thinking • Complexity Reduction • Requirements Translation • Stakeholder Communication

Languages

German — Native

Turkish — Native

English — C1

Education

Software Engineering with Specialization in AI Engineering, Masterschool Institute of Technology

Jan 2026 – Sep 2026 | Remote

- Developed AI and backend applications using Python, FastAPI, Flask, PostgreSQL, REST APIs, SQLAlchemy, and Git/GitHub.
- Built LLM workflows with the OpenAI API, structured outputs, function calling, RAG, and vector search to transform unstructured input into reliable application data.

Diplom-Ingenieur (FH), Automotive Engineering,

Hochschule Karlsruhe – University of Applied Sciences

Sep 2004 – Aug 2011 | Karlsruhe, Germany

- **Diploma Thesis:** *Analysis and Optimization of Manufacturing Processes in a Machinery Company Based on Lean Production Principles* – completed at Kiesling Maschinentechnik GmbH; analyzed manufacturing workflows, identified bottlenecks and waste, and developed process improvements based on Lean Production principles
- Focus areas: Automotive Systems, Production Planning & Control, Quality Management, and Systems and Control Engineering.

Projects

AI Start Map 🌐, Python • FastAPI • PostgreSQL • SQLAlchemy • OpenAI API • Structured Outputs • RAG • FAISS

Jul 2026

- Built a web-based AI diagnostic for solo business owners and micro-businesses that captures current workflows and identifies the three highest-value automation opportunities.
- Developed a multi-stage LLM and RAG pipeline for structured process extraction, targeted follow-up questions, knowledge retrieval, and effort-benefit prioritization.
- Implemented validated structured outputs and server-side guardrails to ensure reliable results and prevent internal knowledge-base references from being exposed to users.

Kompass 🌐, Python • FastAPI • PostgreSQL • Whisper • Twilio • WhatsApp API • Structured Outputs • LLM Pipeline

Jul 2026

- Built a voice-first wellness tracking system on WhatsApp that transforms natural voice messages into structured longitudinal data on mood, energy, sleep, stress, and daily events.
- Developed an end-to-end speech and data pipeline using Twilio, ffmpeg, Whisper, LLM-based structured extraction, FastAPI, and PostgreSQL.
- Implemented historical summaries and pattern analysis to identify recurring relationships and generate personalized follow-up questions while maintaining clear boundaries around diagnostic and treatment claims.

MoviWeb 🌐, Python • Flask • SQLAlchemy • SQLite • OMDb API • Unicorn • Render

May 2026

- Built and deployed a full-stack web application for managing personalized movie collections.
- Integrated the OMDb API and implemented relational data models and CRUD functionality.

Professional Experience

International Living, Parenting & Professional Realignment

Jan 2021 – Dec 2025 | International

- Provided one-to-one mentoring, using structured communication and facilitation to help individuals gain clarity, make decisions, and translate goals into concrete next steps.
- Lived in Türkiye, Mexico, Thailand, and Indonesia, strengthening cross-cultural communication, adaptability, and independence.

Technical Product Specialist / Technical Consultant – Driver-Assistance Systems, Mercedes-Benz AG

Oct 2012 – Dec 2020 | Sindelfingen, Germany

- Owned global technical market support and field-performance monitoring for camera and parking-assistance systems across 50+ markets, 14+ model lines and 30+ vehicle generations
- Investigated complex field issues across 10+ functions and numerous hardware and software variants, tracing cross-system faults involving software releases, control units, gateways and market-specific conditions
- Coordinated engineering, quality, production, compliance, product-liability and market teams to develop corrective measures and repair solutions
- Translated complex field data and risk assessments into decision-ready analyses and field-action recommendations for senior management; represented After-Sales in launch and fault-resolution teams
- Developed technical training and communication concepts and delivered presentations and workshops at international events with 1,000+ international representatives

mindful@Daimler Ambassador (Concurrent Role), Mercedes-Benz AG

Jul 2019 – Jun 2020 | Stuttgart, Germany

- Co-developed the company-wide mindful@Daimler network and ambassador program, creating training and presentation materials and facilitating cross-functional formats focused on communication, collaboration, and knowledge sharing.

Account Manager – Engineering Services, FERCHAU

Nov 2011 – Sep 2012 | Stuttgart, Germany

- Managed Mercedes-Benz as the primary client account, translating requirements from multiple technical departments into tailored proposals and engineering service solutions
- Acquired new business, negotiated contracts and pricing, and coordinated project delivery and engineering staffing
- Led up to 3 engineering professionals, overseeing recruitment, deployment planning, workload allocation and professional development

Early Engineering Experience

Requirements Engineering Intern, MBtech Group

Apr 2009 – Nov 2009 | Sindelfingen, Germany

- Redesigned a complex requirements-documentation structure in IBM Rational DOORS, defining document types, relationships, and attributes; the resulting structure was implemented across several software development projects.

Service Engineering Intern, Mercedes-Benz USA

Mar 2007 – Aug 2007 | Montvale (NJ), USA

- Developed databases and tracking systems for engine-replacement cases and technical priorities, analyzed US product-quality and diesel-service trends, and presented findings to department and executive management.

Product Quality & Technical Analysis Intern, DaimlerChrysler AG

Feb 2006 – Aug 2006 | Sindelfingen, Germany

- Developed a cross-series database matrix for electronic control units and operating panels, conducted warranty and goodwill analyses, and prepared technical evaluations and decision documents.